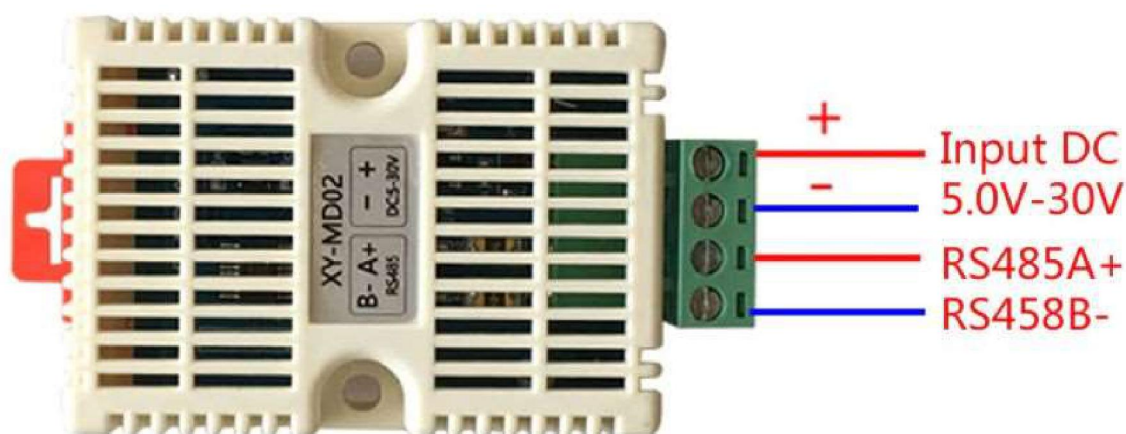


XY-MD02

1. Description:

The product adopts an industrial-grade chip and a high-precision SHT20 temperature and humidity sensor to ensure strong reliability, high accuracy, and excellent interchangeability. It uses an RS485 hardware interface with lightning protection and is fully compatible with the standard industrial Modbus RTU protocol.

This product supports both MODBUS RTU and standard UART communication protocols, allowing users to select the required communication mode. The UART interface supports automatic reporting, and data can be sent automatically through the RS485 serial interface to the connected monitoring tool.



2. Features:

- Supports MODBUS RTU protocol
- RS485 communication distance up to 1000 meters
- Standard DIN35 mounting rail installation
- High precision
- Industrial-grade SHT20 temperature and humidity sensor with RS485 communication
- Supports both Modbus RTU and standard UART communication protocols, and users can select the appropriate mode
- Baud rate can be configured by the user
- UART mode supports automatic upload, and upload speed is configurable by the user

3. Parameters:

- Product Name: Modbus RTU RS485 SHT20 Temperature and Humidity Transmitter
- Product Number: XY-MD02
- Working Voltage: DC 5V ~ 30V
- Output Signal: RS485 signal
- Communication Protocol: Modbus RTU

- Communication Address: 1 to 247 (default 1)
- Temperature Range: -40 to 60 °C
- Temperature Precision: +/-0.5°C
- Temperature Resolution: 0.1°C
- Humidity Range: 0%RH to 80%RH
- Humidity Accuracy: ±3%RH
- Humidity Resolution: 0.1%RH
- Power Consumption: <0.2 W
- Work Temperature: -40 to 85 °C °C
- Working Humidity: 0%RH to 95%RH
- Size: 65×46×28.5mm

Modbus Protocol			
Function Code			
Command Register		Funciotn	
0x03		Read keep register	
0x04		Read input register	
0x06		Write a single keep register	
0x10		Write more keep registers	
Register Type	Register Address	Register Contents	Bytes
Input Register	0x0001	Temperature	2
	0x0002	Humidity	2
Keep Register	0x0101	Device Address	2
	0x0102	Baud Rate: 0:9600 1:14400 2:19200	2
	0x0103	Temperature Correction -10°C~10°C	2
	0x0104	Humidity Correction -10%RH~10%RH	2

Modbus Protocol Communication Format								
Master Send Format								
Device Address	Function Code	Starting Address Hi	Starting Address Li	Quantity Hi	Quantity Li	CRC Hi	CRC Li	
Response Format from Slave								
Device Address	Function Code	Bytes	Register 1 Hi	Register 1 Li	Register N Hi	Register N Li	CRC Hi	CRC Li
ModBus Command								
Master Read Temperature Command Frame(0x04)								
Device Address	Function Code	Starting Address Hi	Starting Address Li	Quantity Hi	Quantity Li	CRC Hi	CRC Li	
0x01	0x04	0x00	0x01	0x00	0x01	0x60	0x0A	
Response Temperature Value from Slave								
Device Address	Function Code	Bytes	Temp Hi	Temp Li	CRC Hi	CRC Li		
0x01	0x04	0x02	0x01	0x31	0x79	0x74		
For example: Temperature value=0x131, converted to a decimal 305 , so the actual temperature value = 305/10 = 30.5°C Note: Temperature is signed hexadecimal number, temperature value = 0xFF33, converted to a decimal - 205, so the actual temperature = -20.5 °C								
Master Read Humidity Command Frame(0x04)								
Device Address	Function Code	Starting Address Hi	Starting Address Li	Quantity Hi	Quantity Li	CRC Hi	CRC Li	
0x01	0x04	0x00	0x02	0x00	0x01	0x90	0x0A	
Response Humidity Value from Slave								
Device Address	Function Code	Bytes	Humidity Hi	Humidity Li	CRC Hi	CRC Li		
0x01	0x04	0x02	0x02	0x22	0xD1	0xBA		
For example: Humidity Value = 0x222, converted to a decimal 546, so actual humidity value = 546/10 = 54.6%RH								
Continuous Read Temperature and Humidity Command Frame(0x04)								
Device Address	Function Code	Starting Address Hi	Starting Address Li	Quantity Hi	Quantity Li	CRC Hi	CRC Li	
0x01	0x04	0x00	0x01	0x00	0x02	0x20	0x0B	
Response Temperature and Humidity Value from Slave								
Device Address	Function Code	Bytes	Temp Hi	Temp Li	Humidity Hi	Humidity Li	CRC Hi	CRC Li
0x01	0x04	0x04	0x01	0x31	0x02	0x22	0x2A	0xCE

Read Keep Register(0x03)							
Read Device Address from Slave							
Device Address	Function Code	Starting Address Hi	Starting Address Li	Quantity Hi	Quantity Li	CRC Hi	CRC Li
0x01	0x03	0x01	0x01	0x00	0x01	0xD4	0x36
Response Data from Slave							
Device Address	Function Code	Bytes	Slave Add Hi	Slave Add Li	CRC Hi	CRC Li	
0x01	0x03	0x02	0x01	0x02	0x30	0x18	
Modify Contents of Registers(0x06)							
Modify Slave Address Register							
Device Address	Function Code	Register Address Hi	Register Address Li	Value Hi	Value Li	CRC Hi	CRC Li
0x01	0x06	0x01	0x01	0x00	0x08	0xD8	0x30
Note:For example,this command is used to change slave address to 0x08.							
Send/Response Data from Slave							
Device Address	Function Code	Register Address Hi	Register Address Li	Value Hi	Value Li	CRC Hi	CRC Li
0x01	0x06	0x01	0x01	0x00	0x08	0xD4	0x0F

Continuously Change Keep Registers(0x10)												
Device Address	Function Code	Starting Address Hi	Starting Address Li	Quantity Hi	Quantity Li	Bytes	Register Address 1 Hi	Register Address 1 Li	Register Address 2 Hi	Register Address 2 Li	CRC Hi	CRC Li
0x01	0x10	0x01	0x01	0x00	0x02	0x04	0x00	0x20	0x25	0x80	0x25	0x09

For example, this command is used to change slave address to 0x20 by register 1. That is 32. Set Baud Rate to 0x2580 by register 2. That is 9600

Response Data from Slave							
Device Address	Function Code	Starting Address Hi	Starting Address Li	Register Num Hi	Register Num Li	CRC Hi	CRC Li
0x01	0x10	0x01	0x01	0x00	0x02	0x11	0xF4

Note:

- 1.This product integrating MODBUS protocol and ordinary UART communication protocol, users can choose communication protocols, UART communication support automatic report function.
2. output temperature and humidity automatically after connect the RS485 serial interface mode tool.

UART Communication Protocol	
Baud Rate	9600
Bit	8
Stop Bit	1
Check Bit	No
Command	Function
READ	Read temperature and humidity
	For example:
	27.4°C,67.7%
	Temperature is 27.4°C Humidity is 67.7%RH
AUTO	Start the temperature and humidity automatically report function (Same as READ)
STOP	Stop the temperature and humidity automatically report function
BR:XXXX	Set baud rate 9600-19200
	For example:
	BR:9600
	Set baud rate to 9600
TC:XX.X	Set the temperature calibration (-10.0~10.0)
	For example:
	TC:02.0
	Set calibration to 2°C
HC:XX.X	Set the humidity calibration (-10.0~10.0)
	For example:
	HC:-05.1
	Set calibration to -5.1%RH
HZ:XXX	Set the temperature and humidity reporting rate. Range is 0.5,1,2,5,10
	For example:
	HZ:2
	Set reporting rate to 2Hz
PARAM	Read system current Set Value
	Return:
	TC:0.0,HC:0.0,BR:9600,HZ:1
	Temperature Calibration:0.0°C
	Humidity Calibration : 0.0%RH
	Baud Rate : 9600
	Report Rate : 1Hz
	SLAVE_ADD:1
	ModBus Slave Address is 0x01